Part 1: Lin Alg $A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$

Q1: $AB: \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ $\begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix}$ $(A+B)$

$BA: \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$ (B^T)

$(A+B)^T: \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}$

$AB^T: \begin{bmatrix} 11 & -1 \\ 4 & -1 \end{bmatrix}$

trace(A): 4

trace(AB): 6

Q2: $M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix}$

$(2-\lambda)^2 - 1 = 0$

$2-\lambda = 1$

$\underline{\lambda = 1}$

$\left(\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - 1\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)\vec{x} = \vec{0}$

$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\vec{x} = \vec{0}$

$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \vec{0}$

$x_1 - x_2 = 0$ $\quad \underline{\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}}$

$x_1 = x_2$

$Mv = \lambda v$

$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Q3: $trace(AB) = trace(BA)$

$tr(AB) = \sum_i \sum_k A_{ik} B_{ki}$ ($i=j$ on diagonal)

$= \sum_k \sum_i B_{ki} A_{ik}$

$= \sum_k (BA)_{kk}$

$= tr(BA)$

Q4: $C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$

$\det(C) = 24 - 14 = 10 \neq 0$

$C \cdot C^{-1} = I$

$\left[\begin{array}{cc|cc} 4 & 7 & 1 & 0 \\ 2 & 6 & 0 & 1 \end{array}\right]$

$\left[\begin{array}{cc|cc} 1 & 7/4 & 1/4 & 0 \\ 2 & 6 & 0 & 1 \end{array}\right]$

$\left[\begin{array}{cc|cc} 1 & 7/4 & 1/4 & 0 \\ 0 & 6 - 7/2 = 5/2 & -1/2 & 1 \end{array}\right]$

$\left[\begin{array}{cc|cc} 1 & 7/4 & 1/4 & 0 \\ 0 & 1 & -1/5 & 2/5 \end{array}\right]$

$\left[\begin{array}{cc|cc} 1 & 0 & 1/4 + 7/20 & -14/20 \\ 0 & 1 & -1/5 & 2/5 \end{array}\right]$ $-7/10$

$\left[\begin{array}{cc|cc} 1 & 0 & 3/5 & -7/10 \\ 0 & 1 & -1/5 & 2/5 \end{array}\right]$

$C^{-1} = \begin{bmatrix} 3/5 & -7/10 \\ -1/5 & 2/5 \end{bmatrix}$

Part 2: Probability, MLE, MAP Est.

Q1:

a) $c\sum_{k=0}^{\infty} \frac{3^k}{k!} = ce^3 = 1$

$\underline{c = \frac{1}{e^3}}$

b) $E[X] = \sum_{k=0}^{\infty} k\left(e^{-3}\frac{3^k}{k!}\right) = e^{-3}\sum_{k=0}^{\infty} k \cdot \frac{3^k}{k(k-1)!}$

$= e^{-3}\sum_{k=1}^{\infty} \frac{3 \cdot 3^{k-1}}{(k-1)!}$ $\quad j = k-1$

$= 3e^{-3}\sum_{j=0}^{\infty} \frac{3^j}{j!}$

$= 3e^{-3} \cdot e^3$

$= \underline{3}$